

Practicamente al final de la introduccion, se proponen los objetivos de este paper:

1. determine the clustering properties of galaxy clusters from state-of-the-art simulations where galaxy properties are derived from semi-analytical modelling
2. test how much the clustering properties evidenced on ideal mock catalogues can be recovered when degrading the redshift information to reproduce the photometric uncertainty on redshift expected in future cluster experiments. (Our objective is to optimize the analysis of cluster clustering from forthcoming cluster catalogues that will be issued from the ongoing and future large multiband photometric surveys.)

1 Secciones

1.1 Introduccion.

- *One of the major challenges in modern cosmology is to explain the observed acceleration of the cosmic expansion, determining if it is due to a positive cosmological constant, a time-varying dark energy component or a modified theory of gravity.*
 - Esta es la excusa para el estudio, es decir le da pie a la necesidad de hacer el trabajo, en este caso el estudio seria util enmarcado en la necesidad de explicar la expansion acelerada del universo.
- Para responder a estas preguntas ellos lo que plantean es utilizar los clusters galacticos:
 - *In fact galaxy cluster counts as a function of redshift and mass are sensitive to dark energy through their dependence on the volume element and on the structure growth rate . One intrinsic difficulty in constraining the cosmological models with galaxy cluster counts comes from uncertainties in cluster mass estimates and on the difficulty to calibrate related mass proxies.*
- Para hacer los analisis anteriores proponen utilizar el estudio de la funcion de correlacion de los clusters.
 - *Cluster clustering can be measured through the twopoint correlation function, the Fourier transform of the power spectrum, which is one of the most successful statistics for analysing clustering processes.*
 - *In cosmology, it is a standard tool to test models of structure formation and evolution. The cluster correlation function is much higher than that of galaxies. This is a consequence of the fact that more massive haloes correspond to higher and rarer density fluctuations, which have a higher correlation amplitude.*
 - *Galaxy clusters are associated to the most massive virialised dark matter haloes, and as a consequence their correlation function is strongly amplified.*
- Ellos quieren utilizar los datos de Sloan, sin embargo hay problemas con la estima de redshifts fotometricos.
 - However, among the several difficulties to be overcome in using clusters as cosmological probes is the impact of photometric redshift errors.

- While some of these surveys have (in general partially) a spectroscopic follow up, many forthcoming large galaxy surveys will have only the photometric information in multiple bands, so that their cluster catalogues will be built on the basis of state-of-the-art photometric redshifts.
- Using those instead of real redshifts will cause a **positional uncertainty** along the line of sight inducing a damping of clustering at small scales and a smearing of the acoustic peak

1.2 Simulaciones

Como se vio anteriormente, ellos quieren utilizar funciones de correlacion para medir el clustering de los clusters. Asi que por ende, deben utilizar alguna simulacion.

We use a public light-cone catalogue constructed using a semi-analytic model of galaxy formation onto the N-body dark matter halo merger trees of the Millennium Simulation, based on a lambda cold dark matter (Λ CDM) cosmological model with the following parameters:

$$\Omega_M, \Omega_\Lambda, \Omega_b, h = 0.25, 0.75, 0.045, 0.73$$

corresponding to the first year results from the Wilkinson Microwave Anisotropy Probe.

The area covered by the light-cone is 500 deg^2 ; the final mock catalogue is magnitude-limited to $H=24$ (to mimic the Euclid completeness) with a maximum redshift at $z=3$.

1.3 Funciones de Correlacion

Aqui se hace un breve repaso de las funciones de correlacion.

$$dP_{12} = n^2 [1 + \xi(r)] dV_1 dV_2$$

Aqui dP_{12} es la probabilidad de encontrar dos objetos en los volumenenes dV_1, dV_2 separados a una distancia r . n es la densidad media mientras que ξ es la funcion de correlacion que mide el exceso de probabilidad de encontrar un par en relacion a una distribucion de Poisson.

El estimador que se utiliza para calcular la funcion de correlacion es el de Landy & Szalay.

$$\xi(r) = \frac{DD(r) - 2DR(r) + RR(r)}{RR(r)}$$

Se dan unos cuantos argumentos para utilizar este estimador:

- less sensitive to the size of the random catalogue and better in handling edge corrections
- Se argumenta ademas el paper de Kerscher 2000 que compara la performance de distintos estimadores para la funcion de correlacion.
- DD : is the number of data-data pairs counted within a spherical shell of radius r and $r+dr$
- DR : refers to the number of data-random pairs
- and RR refers to the random-random pairs

Ahora se indica algo muy importante, que deacuerdo a Peebles, la funcion de correlacion en el espacio real, sigue una Power Law:

$$\xi(r) = \left(\frac{r}{r_0}\right)^{-\gamma}$$

- r_0 : es la longitud de correlacion y γ es la pendiente.
- The random catalogues we use reproduce the cluster redshift **selection function**, estimated by smoothing the cluster redshift distribution through a kernel density estimation method.

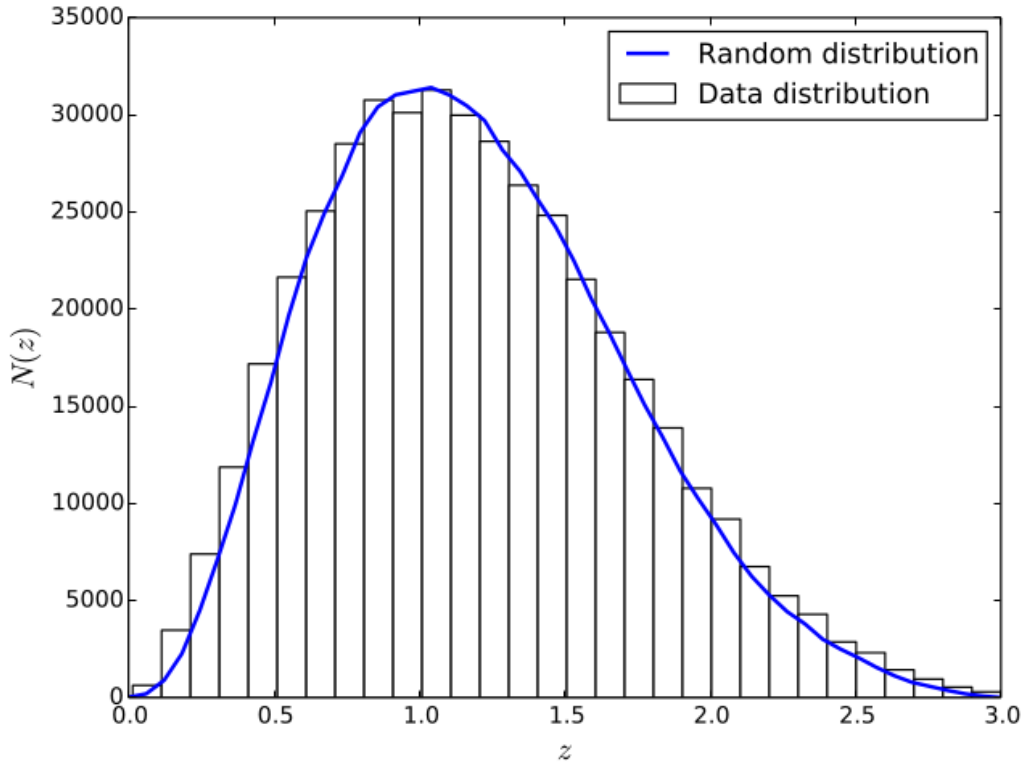


Fig. 1. Redshift distribution of the entire catalogue with $0.0 < z_c < 3.0$. The data distribution is shown as the histogram along with the blue line specifying the distribution of the random catalogue we use for calculating the two-point correlation function.

Figure 1. Distribucion de los redshifts comparadas con su catalogo simulado y los datos.

- Sobre los errores: Errors are calculated from the covariance matrices using the jackknife resampling method

1.4 Estima de la funcion de correlacion con redshifts fotometricos

1.5 Recobrando la funcion de correlacion en el espacio real.

- In the following we will take into account separately the line of sight π and the transverse r_p components of the two-point correlation function.

- Photometric redshifts affect only the line of sight component, introducing an anisotropy in the $\pi-r_p$ plane: the redshift-space correlation function will have a lower amplitude and steeper slope with respect to the real-space correlation function.
- Lo siguiente es del paper de Arnalte (Recovering the real-space correlation function from photometric redshift surveys)[Creo que explica mejor]: The large errors in redshift and hence in the line-of-sight positions produce two effects that have to be considered.
 - On one side, these random shifts in position erase correlations between points, and hence $\xi(r)$ measured according to (1) (Landy Szalay) would be much lower than the real $\xi(r)$.
 - On the other side as the shifts are only in the line-of-sight direction, isotropy of the distribution is lost. Correlation is only lost along the longitudinal direction, but it is conserved in the transverse plane.

- En lineas generales lo que va a hacer ahora es definir las direcciones σ, π . Aqui π esta a lo largo de la linea de la visual. Tipicamente se define como:

$$\mathbf{s} = \mathbf{x}_2 - \mathbf{x}_1 ; \mathbf{l} = \frac{1}{2}(\mathbf{x}_1 + \mathbf{x}_2)$$

- Tras definir estos dos vectores, uno para la separacion espacial entre los objetos: $\mathbf{s}$ mientras que el otro es para denotar la LOS: $\mathbf{l}$.
- Entonces se definen:

$$\pi = \frac{\mathbf{s} \cdot \mathbf{l}}{l} : \text{Componente paralela a LOS}$$

$$\sigma \text{ tiene que ser un escalar del modulo de la componente perpendicular.}$$

Tal y como dijo Diego, es trivial ir desde la funcion de correlacion 3D a la 2D:

$$w_p(\sigma) = \int_{-\infty}^{\infty} \xi(\sigma, \pi) d\pi$$

Esta expresion se reformula como:

$$w_p(r_p) = 2 \int_{\sigma}^{\infty} \xi(r)(r^2 - \sigma^2)^{-1/2}$$

A partir de aca la idea es invertir para encontrar la funcion de correlacion. Debe elegirse un limite finito para trabajar en la realidad.

$$w_p(\sigma, \pi_{\max}) = 2 \int_0^{\pi_{\max}} \xi(\sigma, \pi) d\pi$$

El resultado de la inversion nos daria:

$$\xi(r) = \frac{-1}{\pi} \int_r^{\sigma_{\max}} w'(\sigma)(\sigma^2 - r^2)^{-1/2} d\sigma$$

Observe que hay ahora definidos $\sigma_{\max}$ y $\pi_{\max}$.